4.3 CONCRETE

Learning Objectives

At the end of this lesson you shall be able to

- Know concrete and its ingrediants.
- Understand about water cement ratio.
- List the types of concrete, its uses and properties.
- Know the preparation of cement concrete.
- Understand the precautions to be taken while transportation and placing of concrete.
- Know the grades of concrete.

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4.3.1 Introduction

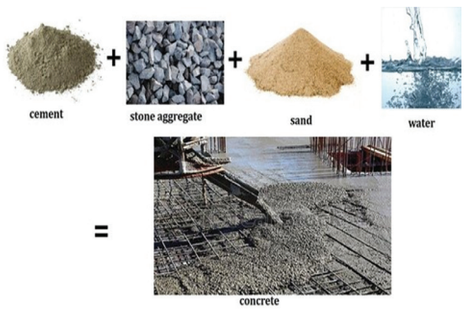
Cement concrete is a major building material used in modern building constructions. It is used in all parts of a building like foundations, superstructure and roofs. It is prepared at site by hand mixing or machine mixing. Nowadays it is also available as a factory made product known as **"Ready Mix Concrete" (RMC)**.

4.3.2 Definition

Concrete is a composition of coarse aggregate, fine aggregate, binding material and water in such proportions that the whole sets into a monolithic mass.

When cement concrete is used without reinforcement, it is called **"Plain Cement Concrete" (PCC)**. If it is reinforced with steel, it is called **"Reinforced Cement Concrete" (RCC)**.

4.3.3 Ingredients of Concrete



The ingredients of concrete are,

1. Binding material (cement or lime)
2. Fine aggregate (river sand or M-sand)
3. Coarse aggregate (broken bricks or broken stone)
4. Water
5. Admixtures in cement

4.3.3.1 Binding Material

Cement or lime are used as the binding material. They bind the individual units of fine aggregate and coarse aggregate

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